~~Title~~Stubbornness and Social Power in Opinion Dynamics: the French–Harary–DeGroot and Friedkin–Johnsen Models

Leonardo Luigi Pepe

November 3, 2026

Abstract

Abstract.

Contents

1 Introduction **2**
1.1 Preliminaries . . . 2
1.1.1 Graphs . . . 2
1.1.2 Matrices . . . 3

2 The French–Harary–DeGroot model **4**
2.1 Introduction to the FDG model . . . 4
2.1.1 The FDG model . . . 4
2.1.2 Consensus and social power . . . 5
2.1.3 Disagreement . . . 6
2.1.4 Rosetta stone . . . 6
2.2 Stubbornness and social power in the FDG model . . . 7
2.2.1 Modelling stubbornness: self-weights and stubborn agents . . . 8
2.2.2 Self-weights, stubbornness and convergence . . . 8
2.2.3 Self-weights, stubbornness and social power . . . 9

3 The Friedkin–Johnsen model **11**
3.1 Introduction to the FJ model . . . 11
3.1.1 The FJ model . . . 11
3.1.2 Convergence and stability . . . 11
3.1.3 How the opinions converge: the matrix V and the augmented graph . . . 14
3.1.4 Influence centrality . . . 16
3.2 Stubbornness and influence centrality in the FJ model . . . 16
3.2.1 Modelling stubbornness: prejudices and stubborn agents . . . 16
3.2.2 Prejudices, stubbornness and convergence . . . 20
3.2.3 Prejudices, stubbornness and influence centrality . . . 25

4 Conclusions **26**

1 Introduction

How do the opinions of a group evolve when its members listen to each other, and what changes when some of them resist? Models of opinion dynamics describe a social network as a weighted digraph, whose nodes are the agents and whose edges carry the influence of one agent on another, and update the opinions by repeated averaging. This report follows the tutorial by Proskurnikov and Tempo [5], the course slides [2, 3, 1] and, as a support, the book by Bullo [4].

The historical account of [5] explains why the report compares two models. The first model of this kind was proposed by French in 1956: every agent replaces its opinion with the mean of the opinions displayed to it, and French used it to measure the *social power* of each agent, i.e. its ability to influence the common opinion of the group; his conditions for consensus were proved rigorously by Harary in 1959 and 1965. In 1974 DeGroot extended the model to an arbitrary stochastic matrix, as a procedure to reach a consensus of subjective probabilities: this is the *French–Harary–DeGroot* (FDG) model. In 1964 Abelson had proposed its continuous-time counterpart and observed that consensus, the outcome of a very broad class of such models, is too strong to be realistic: what must be assumed to explain the persistent disagreement observed in real groups? This question, the *community cleavage problem*, points to agents that resist the others. In 1968 Taylor extended Abelson's model with constant external opinions, the *prejudices*, to which the agents remain attached; its discrete-time counterpart came only in the 1990s, with the *Friedkin–Johnsen* (FJ) model, in which every agent mixes the average of the opinions it listens to with a fixed prejudice, and which has been validated experimentally on small and medium-size groups.

The thread of the report is stubbornness: an agent that never changes its opinion, and the graded mechanisms of which it is the extreme case, a self-weight in the FDG model and a prejudice in the FJ model. We ask how these mechanisms change three things: whether the opinions converge, what they converge to, and how much each agent weighs in the result, measured by the social power in the FDG model and by the influence centrality in the FJ model. The two models are compared on the same small networks, with agent 1 as the agent that resists, so that every difference comes from the model and not from the data. The continuous-time models of Abelson and Taylor are mentioned only as an overview and are not studied in this report.

Section 1.1 recalls the notions on graphs and non-negative matrices used throughout. Section 2 introduces the FDG model and studies stubbornness and social power in it; Section 3 does the same for the FJ model, from its convergence and the augmented graph to the influence centrality; Section 4 compares the two models.

1.1 Preliminaries

This subsection collects the notions on graphs and non-negative matrices used for both models, following the course slides [2, 3, 1].

1.1.1 Graphs

A *weighted digraph* $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ has node set $V = \{1, 2, \dots, n\}$, edge set $E \subseteq V \times V$ and a weight $a_{ij} > 0$ for each edge $(i, j) \in E$; an edge (i, i) is a *self-loop*. Throughout the report the nodes are agents, and an edge (i, j) means that agent i is influenced by agent j. This is the convention of the course slides [1]; the tutorial by Proskurnikov and Tempo draws the edges in the opposite direction [5].

Definition 1.1 (Weighted adjacency matrix [2])**.** Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j)\in E})$ of order n the (weighted) adjacency matrix of G is the $n \times n$ matrix A in which:

(i) for each edge $(i, j) \in E$ the (i, j) entry of A is equal to the edge weight a_{ij},

(ii) all other entries of A are equal to zero.

Definition 1.2 (Connectivity of digraphs [2])**.** A directed graph $G = (V, E)$ is

(i) *strongly connected* if there is a directed walk from any node to any other node;

(ii) *weakly connected* if the undirected version of the graph is connected.